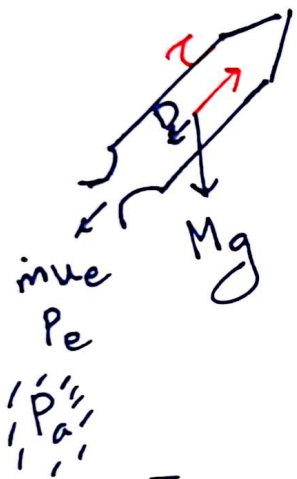


Monday Afternoon 01/13/2025

We review Thrust, u_{eq} & introduce you specific impulse, different mass of the rocket & acceleration Eqⁿ

$$\tau = \dot{m} u_e + (P_e - P_a) A_e = \dot{m} u_{eq}, \text{ where } u_{eq} = u_e + \frac{RT_e}{u_e} \left(1 - \frac{P_a}{P_e}\right)$$

$$I_{sp} : \text{specific impulse} = \frac{u_{eq}}{g_e} = \frac{\dot{m} u_{eq}}{\dot{m} g_e} = \frac{\tau}{\dot{m} g_e}$$



Note: M : mass of the rocket
 $\dot{m}$: mass flow rate of exhaust gas

$$[I_{sp}] = \frac{\left[\frac{m}{s}\right]}{\left[\frac{m}{s^2}\right]} = [s]$$

I_{sp} : How powerful is "effective exhaust velocity" compared to "gravity acceleration"
or How much thrust is produced per weight of exhaust gas

Range of I_{sp}

chemical Rocket: $200s < I_{sp} < 450s$: high $\dot{m} \rightarrow$ low I_{sp}

nuclear " $I_{sp} \sim 900s$

electric " $I_{sp} \sim 2000s$: low $\dot{m} \rightarrow$ high I_{sp}

(P-11)

Falcon 9 Rocket $\rightarrow \begin{cases} I_{sp}(s.l) = 280s \\ I_{sp}(vac) = 350s \end{cases}$

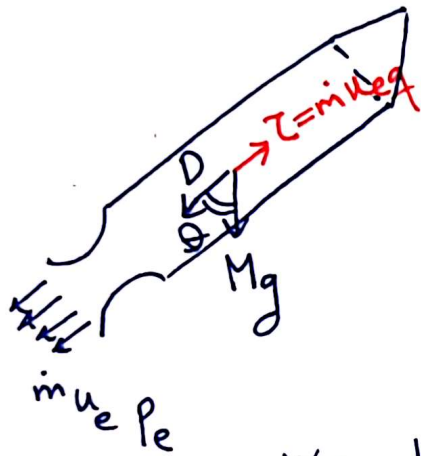
$$I_{sp} = \frac{u_{eq}}{g_e}$$

① $u_{eq} = u_e + \frac{1 - \frac{P_a}{P_e}}{\rho_e} \quad @ \text{vac } P_a = 0 \Rightarrow$

② low g at high altitude $\frac{1 - \frac{P_a}{P_e}}{\rho_e} = 1$

"Vehicle Acceleration"

M : Mass of the rocket



$$\frac{dM}{dt} = -\dot{m} \rightarrow \text{mass flow rate of exhaust gas}$$

$$\rightarrow M - M_0 = -\dot{m}t \Rightarrow M = \underbrace{M_0}_{\text{initial mass}} - \underbrace{\dot{m}t}_{\text{Mass at time } t}$$

we define burnout mass: $M_b = M_0 - \dot{m}t_b$

burnout: when all propellant burn!
fuel + oxidizer

Mass of propellant $\rightarrow M_p = \dot{m}t_b \Rightarrow M_b = M_0 - M_p$

$R = \text{mass ratio} = \frac{M_0}{M_b} \begin{cases} R \text{ small } (M_b \uparrow) \text{ is good as } M_p \text{ is low} \\ R \text{ large } (M_b \downarrow) \text{ is bad as } M_p \text{ is high} \end{cases}$

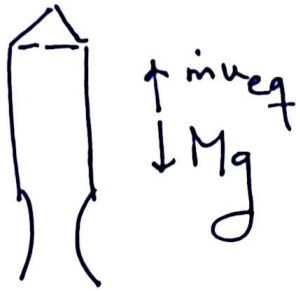
(P-2)

Let's write momentum Eqⁿ for rocket (Newton 2nd law)

$$\bar{F} = M a = M \frac{du}{dt} = \dot{m} u_{eq} - Mg \cos \theta - D \quad \div M$$

$$a = \frac{du}{dt} = \frac{\dot{m} u_{eq}}{M} - g \cos \theta - \frac{D}{M} \quad \text{Acceleration Eqⁿ}$$

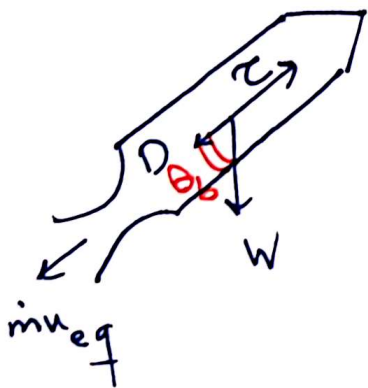
consider launch acceleration ($\theta = 0^\circ$ & assume $D = 0$)



$$a_0 = \frac{\dot{m} u_{eq}}{M_0} - g \quad \text{Earth}$$

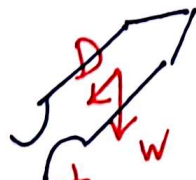
consider burnout acceleration

1- an epsilon second before propellant burnout (yet we have some exhaust)



$$a_{b^-} = \frac{\dot{m} u_{eq}}{M_{b^-}} - g \cos \theta_{b^-} - \frac{D_{b^-}}{M_{b^-}}$$

2- An epsilon second after propellant burnout (no exhaust, no thrust)



(P-3)

No exhaust

$$a_{b^+} = -g \cos \theta_{b^+} - \frac{D_{b^+}}{M_{b^+}}$$

a_{b^+} is a negative term $\rightarrow$ Deceleration

compare a & a_{b^-} , $M_0 \gg M_{b^-} \Rightarrow a \ll a_{b^-}$

" " " " , $g_e > g \cos \theta_{b^-} \Rightarrow a < a_{b^-}$

Note that: $M_b = M_0 - M_p$, M_p is 91% of M_0 for moon rocket, M_p is 86% of M_0 for space shuttle

So you can imagine how large is a_{b^-} compared to a .

$$M_p = \underbrace{\dot{m}}_{\text{fixed}} \underbrace{t_b}_{\substack{\text{large} \\ \text{small}}}$$

a_{b^-} depends on $\dot{m}$, so a_{b^-} increases

$$M_p = \underbrace{\dot{m}}_{\text{fixed}} \underbrace{(t_b)}_{\substack{\text{small} \\ \text{large}}} \longrightarrow a \text{ decreases (small } \dot{m})$$